Food traceability from field to plate

Linus U. Opara and François Mazaud

Linus Opara is with the Centre for Postharvest & Refrigeration Research, Institute of Technology and Engineering, Massey University, Private Bag 11-222, Palmerston North, New Zealand. Tel: +64 6 3505106. Fax: +64 6 3505604. E-mail: L.U.Opara@massey.ac.nz. François Mazaud is with the Agro-Industries and Post-Harvest Management Service (AGSI), Food and Agricultural Organization of the United Nations (FAO), 0100 Rome, Italy. E-mail: Francois.Mazaud@fao.org.

Several factors have led to intensified public scrutiny of the human food supply chain. Consumer concerns for food safety, animal welfare, and the enviro nmental and ecological impact of food production and agro-processing have become increasingly important. These concerns have been exacerbated by several factors, including the trend towards further globalization of the food supply chain, the incidence of new and emerging safety hazards such as the human form of BSE (mad cow disease), and illnesses and deaths resulting from contamination of fresh and processed food. As a consequence of these growing concerns, consumers and other stakeholders in agroindustry now demand transparency in the way food is grown and handled throughout the supply chain, resulting in the emergence of 'traceability' as an important policy issue in food quality and safety. This paper provides a global overview of 'traceability' as a quality index in food trade, and discusses some of the drivers in both developed and developing countries. Policy changes are necessary specifically to incorporate traceability into existing food safety regulations and trade agreements. This will require further investments in information technology for data capture, storage and retrieval. Small-scale farmers in many developing regions moving towards market orientation face considerable technical and financial challenges in implementing appropriate food traceability systems in order to meet marketing compliance requirements.

Agriculture is diverse, multifaceted and multifunctional. It supplies the basic raw material needs for human food, feed and fibre, and plays a crucial role in maintaining the biodiversity of plant and animal species. Agriculture also contributes to the creation of a pleasant and aesthetic physical environment for human habitation and enjoyment. On the other hand, agricultural production and post-harvest practices have also been implicated in negative consequences on humans and the ecology. The incidence of food-related illnesses has been linked to microbial hazards, which enter the food supply chain during growing, handling and processing. The development of the human form of bovine spongiform encephalopathy (BSE) and increasing incidence of food poisoning and other related illnesses linked to the consumption of fresh or processed fruit and vegetables are recent examples.

The negative impacts of agricultural practices on the ecology and environment are also well documented in scientific, policy and public fora. For instance, it has been estimated that for every \$1 of agricultural pesticides used (to produce \$4 of food) the environmental and public health impacts total \$2.[1] Excessive soil degradation, the depletion of natural habitats, and air pollution are some other notable examples. Several technical and sociocultural strategies have been implemented and many more are still under development to assist in mitigating these negative impacts of agriculture.

With respect to food, there is a considerable amount of literature on measuring the benefits and costs of food safety,[2] including the cost of illness and death, and the willingness to pay for safer food. In the USA, for instance, estimates of the total cost of illnesses related to foodborne diseases range from $5–$10 billion per annum, but some studies have reported higher values in the range of $20–$30 billion annually.[3] A recent article by Gregory[4] provides an extensive review of some of the food scares that have occurred during the past 40 years, and considers food buyers' attitudes towards food safety, environmental issues, animal welfare, organic foods and genetically modified foods. The decline in public confidence was associated with an increase in the frequency of food scares, particularly due to outbreaks of BSE and microbial contaminants.

Consumers and other stakeholders in the food supply chain now demand assurances on the quality and safety of food products, as well assurances of minimal impact on the environment and ecology. A good quality assurance system must therefore include evidence of transparency on the way we practise agriculture, including the handling and transformation of the product from farm to plate. To meet these requirements, both growers and post-harvest handlers must demonstrate the traceability of their products.

This article presents a global overview of food traceability, which is gaining considerable importance in agriculture and the food trade. The specific objectives are to:

(1) discuss the competitive and dynamic global environment in which agriculture must operate to produce traceable quality food products;
(2) define traceability as a new index of food quality and safety; and
(3) examine the drivers of food traceability in both developed and developing countries, highlighting the implications for food policy.

Agriculture in a competitive global environment

Although the contribution of agriculture to national economies continues to decline in most developed countries, its importance in food security and peace cannot be overemphasized. The situation is even more obvious in developing countries where most of the people live in rural communities as small-scale subsistence farmers. While less than 10% of the population in most developed economies engage in agriculture, over 70% of those living in developing countries still live in rural areas and rely on agriculture for their livelihood.

Several ongoing trends in the competitive environment for the supply of food and other agricultural products have led to this interest in traceability. The first is related to the declining terms of trade affecting agriculture and the increasing competition among food product suppliers. Many important food sources are oversupplied on a global scale, thereby reducing prices. Two centuries ago, Karl Marx recognized that as economic development occurred the value of agricultural products fell steadily as a proportion of both weekly spending at household level and gross domestic product (GDP) at the national level.[5] Given that the price of agricultural inputs tends to keep pace with economic growth while the returns to agriculture do not, it is expected that agriculture will continue to experience declining terms of trade in the foreseeable future.

Increasing globalization of the world economy is another trend that is radically changing the way we practise agriculture and the way we trade in agricultural products and services. Globalization is characterized by competition and promotes free trade, which in turn demand global harmony in standards of food quality and safety. Among the key imperatives for globalization are the improvement of market access and the reduction of trade barriers. Ideally, these should facilitate the import and export of agricultural products across nations and regions, and between developed and developing countries. Another feature of globalization is the increasing emergence of multinational corporations in the food sector, which are capable of procuring, distributing and selling their products internationally. These corporations continue to form strategic alliances with their former competitors and major producers in different parts of the world, thereby increasing the need to institute reliable quality assurance programmes to manage variability in product quality and minimize the incidence of food safety hazards.

Globalization is not a new phenomenon. In terms of the ratio of international trade to world output,[6] it was in fact more intense in the last quarter of the nineteenth century. However, what is strikingly innovative and pervasive about globalization in the twenty-first century is that it is being accelerated by tremendous ongoing advancements in information and communication technology. For instance, one of the consequences of the expanding application of e-commerce in agriculture is that many commodities, including fresh fruit and vegetables, are traded in a 24-hour cycle around the globe, linking suppliers in remote areas with buyers in major industrialized centres through the Internet. The availability and access to information technology in rural agricultural communities is therefore an important factor in improving the product quality standards and quality assurance systems (including traceability) necessary to participate in the global food trade. Overall, the complexity of the global competitive environment requires that food growers, post-harvest handlers, agro-processors and food safety agencies must all collaborate and work harmoniously to generate, manage and deliver the information necessary to assure the transparency and traceability of the food supply chain.

Trends in patterns of food consumption

The international food trade expanded tremendously towards the end of the last century, enabling consumers in various parts of the world to enjoy a wide range of temperate, tropical and subtropical food products. Changes in food consumption patterns and rising incomes, particularly in the developed world, have contributed to this trend. Increasingly, consumers are eating to express their personal values, concerns and aspirations.

Consumers also want assurances on freshness, naturalness, taste, safety, traceability, health and nutritional benefits, good animal welfare practices, and sustainability of growing and handling practices, which also have minimal negative impacts and ameliorative effects on the environment, zero waste and fair trade. These numerous factors contribute to the emergence of new market-pull factors, which synergistically influence the overall perception of food quality by the consumer. More than ever before, the quality and safety of food has come under considerable scrutiny by consumers and the general public.

Increasing mobility of labour and demographic shifts in major metropolitan areas have resulted in the introduction of new food products and changing food patterns, particularly where a large proportion of new immigrants settle in close communities. With improved access to their 'traditional' foods, first through imports and later through the introduction of new plant and animal material, these new migrants introduce new types of food and cuisine to their new communities.

The rise in fresh food consumption is one of the top 10 mega-trends in food consumption patterns in the USA[7] and in many other developed countries. Particularly critical is the popularity of 'fresh' as the most desirable food label (ranked #5). The quality and variety of fresh fruit and vegetables and fresh meat ranked second and third, right behind a clean store, as the reasons why consumers elected to shop in a grocery store. This study further noted that the emerging trend for organic foods was yet to reach its full potential.

The rapid emergence of 'organic' as a new and important food quality attribute is now widely recognized and documented.[8] For some consumers, 'organic' equates with 'naturalness' or being 'fresh from the farm', and represents a critical influence on the decision to buy. The increasing demand for organic products and the premium prices they command in the marketplace are very significant trends that cannot be ignored by growers and marketers of fresh foods. On a global scale, this demand is currently growing at about 20% annually, and although organic foods account for less than 1% of all food sales in most developed countries, the share of organic products in the food supply system is projected to increase, particularly amongst affluent consumers.

Increasing concerns about food safety, wholesomeness and genetic modification are converting erstwhile sceptics into believers in organic fruit, vegetables and meat products. In the USA, a recent survey showed that 60% of all Americans were either consumers of organic foods or were interested in these products.[9] Within the European Union (EU), consumer demand and awareness of organic products is much higher and growing. Many supermarkets have responded to the increasing consumer demand for information on the food production and handling system by dedicating a proportion of shelf space or product mix to organically grown products.

The EU has introduced regulations to increase food traceability in an effort to win back consumer trust in Europe, which has been undermined by several food scares, such as BSE, in the past few years. For instance, it is now mandatory to issue each product derived from cows (meat cut, minced meat) with an 'identity card', which states where it has come from, who farmed it and where it was slaughtered and butchered.[10] Under this regulation, consumers will know exactly where the meat on their plate has come from and wherever it went during its life. The two-phase implementation process started in September 2000, and stipulates that consumers will be informed of the origins of the animal and where it was butchered. A code number will also be shown on minced meat stating which farm the animal comes from. Implementation of the second phase will be completed by January 2002, when consumers will further be informed of the country in which the cow was born, where it was raised and fed, and where it went before being killed, both for steaks and minced meat. Similar regulations have been adopted in some non-European countries, and in New Zealand, for instance, it became mandatory in June 2000 to ensure that animals in the food chain had 'traceable' numbers.[11]

The application of modern biotechnology in agriculture, particularly for genetically modified (GM) foods, has received considerable interest from the general public as well as from scientists and government. Strong scientific and emotional criticisms of the potential impact of biotechnology on food safety, the environment and ecology of plant and animal species have created a backlash despite the potential benefits for increasing farm productivity. Many development agents also question the relevance of agricultural biotechnology in improving food security and alleviating poverty in developing countries where the greatest and most urgent needs exist for improving agricultural productivity and efficiency. Small-scale subsistence farmers in these regions cannot afford agricultural biotechnology. They also lack the enabling infrastructure and other inputs necessary for proper exploitation of modern biotechnology based on genetic modification of plants and animals. Given the huge resource (finance, technology) gap between resource-poor farmers in developing countries and their counterparts in developed countries, hastily unregulated application of agricultural biotechnology in all regions has the potential to exacerbate the growing partition between the rich North and poor South!

Some food handlers and processors have publicly disassociated their businesses from GM products. While the debate goes on, many new experimental field trials have been halted or cancelled. Many food products already in the market have been recalled and withdrawn, and governments have been forced to establish new food quality regulatory agencies and standards.

A recent study in the UK on consumers' greatest concerns over food showed that food hygiene standards, GM, and the use of chemicals, pesticides and additives were at the top of the list.[12] The study further demonstrated that there was an overwhelming feeling on the part of consumers that they did not have all the information and knowledge, and that there was much confusion about the facts. The combination of these trends in food production, consumption patterns and attitudes, coupled with the overwhelming desire for convenience (eg for minimally processed, ready-to-cook, ready-to-eat food) and environmental protection provides a powerful driving force for increasing consumer demand for quality assurance and traceability in the international food trade.

Dimensions and orientations of food quality

Quality is not a single, recognizable characteristic. It is a dynamic concept, which provides a platform for interaction between producers and other players in the supply chain in order to meet consumer requirements and expectations. The exploitation of superior product quality as a competitive weapon is widely recognized and documented.[13] Although the quality of a food product (or service) is recognized as important for successful agribusiness, there is often a general lack of understanding or agreement on the meaning. This has resulted in a plethora of definitions of quality, often reflecting the type of product, stage in the post-harvest handling chain, or the intended use.[14]

What is needed is a synthesis of the various definitions and approaches to quality, based upon a careful identification of the dimensions and orientations of quality. Garvin[15] proposed eight dimensions of quality based on industrial products, stressing that each dimension was self-contained and distinct, since a product could be ranked high in one dimension while being low in another. Table 1 shows a revised adaptation of Garvin's dimensions of product quality applied to fresh produce.

Table 1. Dimensions of food quality.

Dimension	Explanation	Possible application to the food industry
Performance	Primary product characteristics	Texture, size, appearance, acidity, sweetness
Features	'Bells and whistles'	Packaging, labelling, information on 'how to use', 'alternative uses'
Reliability	Frequency of failure	Regular supply
Conformance	Match with specifications	Minimum variability in product quality standards
Durability	Product life	Storage and shelf life
Aesthetics	'Fits and finishes'	Presentation, display
Perceived quality	Reputation and intangibles	Production system (eg organic *versus* conventional, integrated pest management (IPM) *versus* intensive agri-chemicals), environmental impact, use of genetically modified organisms
Serviceability	Speed of repair	Traceability, speed of recall or replacement

Note: Revised from L.U. Opara, 'New market-pull factors influencing perceptions of quality in agribusiness marketing (or quality assurance for whom?)' in Highley *et al*, eds, *ACIAR Publication No. 100*, Australian Center for International Agricultural Research, 2000, pp 244–252.

A key message in Table 1 is that each dimension of quality imposes different demands on the business, and the implication is that growers and food companies must define the dimensions of quality on which they hope to compete, and should then focus their human and capital resources on these elements. An organization is likely to be more successful in pursuing a strategy of high product quality if it selects a small number of dimensions on which to compete, and then tailors them closely to the needs of its chosen market. These principles apply equally to food quality and trade.

The food supply chain (from production to the consumer) is increasingly becoming a complex web, resulting in mutual relationships and interdependence of growers, post-harvest handlers, processors, marketers and consumers. Because of the variety of players in the supply chain, as well as the potential for different uses of the product at various stages in the chain, different orientations of the 'quality' concept can arise. Producers, researchers and handlers are mostly product-oriented, in that quality is described by specific measurable attributes of the food material (such as texture, flavour, acidity), while consumers, marketers and economists are more likely to be consumer-oriented, in that quality is described by the amalgamation of consumer wants and needs.

Food quality comprises all of the attributes, characteristics and features of a product that the buyer, purchaser, consumer or user expects in order to meet the intended use. A product with excellent quality clearly meets the buyer's or user's highest expectations. Kruithof and Ryall[16] gave a concise definition of quality that is relevant to food trade: 'Quality is consistently meeting the continuously negotiated expectations of the customer and other stakeholders in a way that represents value for all involved.'

These expectations include supplying the right product, at the right time, at the right price, and with the right support service. Therefore, food quality is determined by the relative values of several characteristics, which when considered together, will determine the acceptability of the product to the buyer and ultimately to the consumer.

Developing a good understanding of quality dimensions and orientations is necessary before food marketing can be adequately addressed. The orientation of quality adopted will undoubtedly influence the quality standards (product specifications) and the techniques used to assess them. Quality orientation could also create barriers to quality improvement by focusing on those quality elements that have little relevance to the customer. When considered together, these quality dimensions and orientations create push–pull factors that influence the consumer's perception of quality.[17] To meet the challenges posed by these competitive global trends and by the increasing influence of product quality and safety in food trade, there is a need for a paradigm shift from technology-push (supply-driven) to market-pull (demand-driven) management of food quality. Such a paradigm shift requires the implementation of demonstrable product trace-back programmes as part of an overall quality assurance system.

The catalysts for food traceability

The recent incidents of health hazards and scares in the food supply chain have heightened public concern over the way we practise agriculture, as well as the way we handle and process food.[18] Most notable is the potential link between BSE in cattle and the new variant Creutzfeldt-Jakob disease (CJD) in humans, for which there is currently no cure. Reports of food poisoning incidents and death due to *Salmonella* in poultry and eggs, *Escherichia coli 0157* (*E. coli*) contamination of meat and meat products, fresh, minimally processed and processed fruit and vegetables, dioxins in chicken, and the occurrence of other emerging food pathogens have reduced consumer confidence in the safety of our food systems. The preponderance of these incidents has also resulted in increasing scrutiny of the ability of governments to protect public safety and the capacity of science to understand the nature of some of these food quality and safety problems.[19] The situation is further exacerbated by the release of food products that have been grown from genetically modified materials.

As a result, growers and other practitioners in the post-harvest industry are increasingly subjected to greater scrutiny of their production and post-harvest handling practices, and consumers are increasingly demanding greater 'assurance', 'transparency' and 'traceability' in the food supply chain. The ongoing crisis of confidence in the quality and safety of our food supply chain has resulted in the implementation of a number of quality management interventions to reassure consumers and the general public. In brief, quality assurances must now address diverse issues ranging from farming practices, animal welfare, environmental impacts, to the lifestyles and individual aspirations of selected consumer and/or interest groups. Even when traditional food quality assurance systems are implemented, customers (including consumers) further demand that such systems must permit the traceability of the product along the supply chain.

What is food traceability?

Traceability simply refers to the collection, documentation, maintenance and application of information related to all processes in the supply chain in a manner that provides a guarantee to the consumer on the origin and life history of a product. For a food product, traceability represents the ability to identify the farm where it was grown and sources of input materials, as well as the ability to track the post-harvest history and identify the specific location in the supply chain by means of records. It is more than the 'transparency' of the growing and handling processes or the implementation of standard quality assurance systems (such as Hazard Analysis Critical Control Points — HACCP), although both elements are essential in a traceable food supply system. Traceability adds value to the other quality and safety management strategies by providing the communication linkage for identifying, verifying and isolating sources of non-compliance to agreed standards and customer expectations.

The International Standards Organization system of standards (ISO 9000) stipulates that 'Where and to what extent traceability is a specified requirement, establish and maintain documented procedures for the unique identification of individual products and batches. This identification should be recorded' (ISO 9002, 4.8 — Traceability). Two elements of traceability are implicit in these definitions: (1) identification of the product, and (2) the ability to trace through the records. A good identification system for a product must be unique, legible, resistant to damage, easy to 'capture' for records, tamper-proof and incapable of reuse.[20] The recording procedure must cover the entire supply chain, and this entails identifying, capturing and retaining information in a readily accessible and retrievable form at each step in the supply chain.

Once collected, the records must be retained up to the statutory period of liability for the quality and safety of the product. Due to the interrelatedness and interdependence of the various post-harvest handling steps and processes in determining the final product quality and safety, an effective traceability system relies on the cooperation of all participants in the food supply chain. Accurate records must be kept in accessible forms describing the status of the product and the processes to which it has been subjected at each step in the chain.

Traceability connects and identifies all those involved in the supply chain with responsibility for ensuring the quality and safety of food.[21] It thus enables the identification of the product at each step in the food chain, and therefore provides a valuable index for the diligent grower or post-harvest operator. When incorporated into standard quality assurance techniques such as the HACCP system, traceability enables the food handler to *demonstrate* that the product will meet the stated quality standards, as well as allowing potential source(s) of food contamination to be traced. Traceability is therefore consumer- (market)-driven; it is a complementary tool for food quality management, which is a necessary condition to be fulfilled in ensuring the effective functioning of food safety management systems.

Traceability is fundamentally a proactive approach to food quality and safety management, which advocates preventing the introduction of new and potentially harmful hazards into the supply chain. It complements quality control measures by facilitating the identification and isolation of hazards and the implementation of effective corrective actions in the event of an incident. Therefore, like point inspection and product testing, traceability by itself cannot introduce safety into the product and food handling process. When applied in isolation, it is not a sufficient condition to satisfy the safety requirements of the food chain. However, it assists in preventing the incidence of food safety hazards, and also in decreasing the magnitude of such incidents when they occur by identifying the product affected, what occurred, when it occurred, and who is responsible. When implemented properly as part of an overall quality assurance system, traceability offers a multitude of benefits that improve business efficiency (see Box 1).

Box 1. Benefits and applications of traceability

- ✧ Assurance and reassurance of customers and consumers
- ✧ Regulatory compliance
- ✧ Demonstrable integrity of food supply chain
- ✧ Minimization of and transfer of risks
- ✧ Promotes best allocation of responsibilities
- ✧ Facilitation of internal controls
- ✧ Used to validate and resolve complaints
- ✧ When hazards occur, traceability facilitates effective product recall

How do you trace food in the supply chain?

Due to the potential for litigation and substantial compensation arising from a breach of food quality and safety regulations, careful attention must be paid to implementing a good traceability programme as part of a comprehensive quality management system. At the practical level, traceability basically involves documenting the history (origin, steps and processes) of a product using a recognized and acceptable quality assurance programme such as the HACCP system.[22] This requires decisions on the *reason(s)* for traceability, *what* must be traced, and *how* the trace-back should be carried out.

The process of determining the 'why and what' to trace is accomplished through a detailed hazard analysis (or risk assessment) of all the steps and processes so as to identify those critical points (potential sources of hazard) to food quality and safety. These processes can be integrated into the hazard analysis (HA) component of the HACCP system.[23] HACCP is an internationally recognized tool for effective management of food safety. It is not an actual safety assurance procedure for specific product(s), but is intended for use by individual food companies as a protocol or platform for the development of unique safety assurance procedures to meet their needs.

A basic requirement for a food quality and safety assurance system that includes traceability is that the products must be adequately labelled (eg bar-coded) to facilitate rapid identification of the product and subsequent retrieval of the information previously recorded. In the food trade, typical information that may be stored in a bar code includes the country/region of origin, variety, grower, growing method, harvest date, special treatments, date stored and storage conditions (temperature, relative humidity, air composition), quality specifications, batch number, product destination and nutritional composition. Some of these data may be displayed on a label for easy identification, while the rest can be retrieved from storage.

Many factors must be taken into consideration in establishing a traceability system, including: customer (wholesaler, middleman, retailer, consumer) requirements, regulatory standards, product (type, attributes), type of handling and/or processing, distribution and marketing channels, and the availability and access to technology for labelling, marketing and data recording. A basic roadmap is useful in implementing a functional traceability system (see Box 2).

Part of the documentation used for traceability should include a flow chart of the processes in the supply chain from production to plate (see Figure 1). This pictorial form of documentation facilitates the drafting of other supporting documents. It also makes it easier to understand, apply and demonstrate the flow of the food product through the chain. Using such forms of documentation, the traceability system could be audited through experimental trials for validation and verification. Periodic and independent verification by authorized external bodies is essential to ensure the credibility and effectiveness of food traceability programmes.

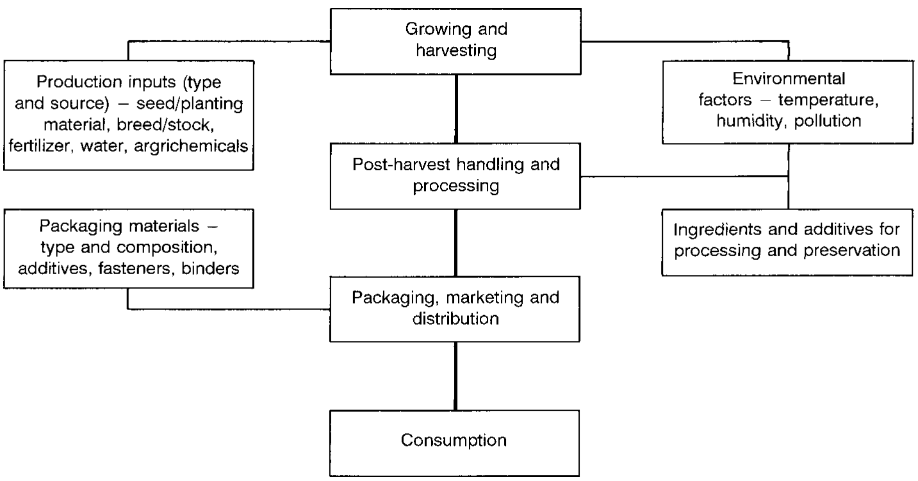

Figure 1. Stages in the food supply chain and type of data recorded to facilitate traceability. Transportation is an important post-harvest operation that occurs between the various main stages in the food supply chain. The collection and documentation of verifiable data (see rounded rectangles) on inputs and process conditions during each stage in the supply chain is a key element in successful trace-back and quality assurance.

Box 2. Basic roadmap of food traceability

(1) Draw a flow chart of the product supply chain, from farm to plate and including links to sources of material inputs such as seed, feed and ingredients.
(2) Appoint a quality assurance officer who is also responsible for product traceability.
(3) Conduct a hazard analysis of the supply chain (using HACCP principles as part of the overall quality assurance system).
(4) Document the reasons for embarking on the traceability of your product.
(5) Write down what information/data must be recorded and traced back at each step in the supply chain.
(6) Specify who will be responsible for collecting and recording the data.
(7) Develop a unique labelling system or bar code for easy identification of the product.
(8) Document how the trace-back is to be carried out (include a trace-back diagram).
(9) Test, validate and verify your traceability system.
(10) Document all decisions and actions.

Misuse and abuse of traceability

We would like to emphasize that traceability is not a mere 'name tag' or 'ear tag' commonly used in the livestock business. It is more than a 'label' that merely identifies the type of product, its origin and the price. Alhough accurate and adequate labelling is essential, the effectiveness of any traceability system is dependent on good record keeping, which facilitates trace-back.

Traceability can be a double-edged sword. Its beneficial role in promoting consumer confidence in the food chain and assistance in settling disputes arising from a breach of food safety regulations and contractual obligations has been noted earlier. However, traceability can also be used to perpetrate negative influences in food trade. Growers, post-harvest operators, consumers, regulatory agencies and government all need to be aware and guard against such misuse and abuse of traceability so that the integrity of the product, process and post-harvest handlers is protected.

Adequately identified and labelled food products can be subject to 'hidden' trade barriers against competing local products in retail stores, particularly if consumers perceive unfair competition against local (home-made) products. In extreme cases, this can result in 'unofficial' boycott and sabotage of the imported product. The sentiments that foster such acts are usually transmitted by word of mouth within the community or by vocal interest groups.

Effective traceability requires considerable financial resourcing and technological inputs. Therefore, unfair and excessive demands for traceability can impede free trade, particularly to the disadvantage of resource-poor farmers in developing countries. Efforts must therefore be made to develop dialogue between trading partners so as to assure the protection of the consumer and facilitate an improved flow of goods and services.

Drivers of traceability in developed and developing countries

Growing interest in food traceability is linked to several related factors that have emerged mainly in developed countries. Foremost is the recent crisis of confidence in the food system as a consequence of (1) the rising incidence of illnesses and deaths attributable to biological hazards in fresh and processed foods, and (2) the unresolved debate and tension arising from the introduction of genetically modified materials into human food.

Debate on the potential impacts of genetic modification on the safety of human food systems and the environment has been hotly contested among scientists and the general public, and this debate is bound to linger for much longer. Many influential lobbies and interest groups (such as the environmentalists) have invested considerable resources to campaign against the development and application of such technology in agriculture. From a government and productivity perspective, illnesses and deaths due to food-borne hazards represent a major cost to the public health systems, and contribute to low productivity of labour. It is in the interests of both government and the private sector, therefore, to reduce the incidence of such illnesses and deaths.

The interest of both the general public and food producers in traceability has been heightened by the food product liability system.[24] Furthermore, current legal incentives to produce safer food are slightly stronger in outbreak situations and in markets where food-borne illnesses can be easily traced to individual farms. For product liability cases, the three causes of action are strict product liability (defect in product), negligence (grower's/manufacturer's/seller's conduct) and breach of warranty (whether the product conforms to warranty/specifications). However, it is worth noting that the loss of consumer confidence as a result of poor quality and unsafe food products can result in unprofitable business, particularly when the source(s) of food hazard cannot be traced. In a study of 294 cases of food product liability, Buzby and Frenzen[25] found that firms paid compensation in 56% of the cases. Although the legal incentives to produce safer food are weak in most countries, small and medium-scale enterprises have very limited capacity for the legal fees and compensation that could result from litigation.

The globalization of a diverse world food supply system is another driver for traceability. Growers in developed countries have also been vocal in demanding assurances that imported foods and raw materials would not damage domestic agricultural and ecological systems. With increasing consumption of food products grown in locations far removed from the point of production, most developed countries have taken measures to protect the quality and safety of fresh and processed food imports (and invariably the health and safety of citizens). Many countries in Europe and elsewhere have set up national food standards authorities to regulate food quality and safety. In 1996, President Clinton brought into effect the Food Quality Protection Act in the USA. On 2 October the following year, he announced a new initiative to ensure the safety of imported and domestic fresh fruits and vegetables, and charged the Food and Drug Administration and the USDA with developing voluntary industry guidelines. As a result, regulations and guidelines have been issued to minimize microbial food safety hazards for fresh fruits and vegetables, covering the growing, packing, processing and distribution of most fruits and vegetables that are destined for the US market.

Related to globalization is the increasing influence of large multinational and transnational food corporations, which command a high share of the international food trade. With materials sourced from diverse growing regions, often several thousand kilometres apart, the need to minimize product variability and ensure food safety becomes imperative. Traceability provides the tools and information for ensuring that food products can be identified and monitored.

International rejection and condemnation of slavery and exploitation of child labour have put pressure on organizations and individuals to stop these practices and adopt equitable and sensible labour management practices. For food companies operating in regions where these practices occur, evidence of traceability in their supply chain that includes data on labour policy and practices could be a valuable tool in reassuring their customers and the general public.

Rapid advances in information and communication technology constitute another set of drivers for traceability. These include new technologies for data capture, processing, storage and retrieval. With their increasing availability and declining cost, these technologies can be economically incorporated into food quality and safety management programmes.

The situation is somewhat different in many developing countries, even though the farmers there also face the same unfavourable terms of trade generally affecting agriculture. To date, agriculture is still small-scale and predominantly at subsistence level in these countries. Given that the individual small-scale farmer cannot compete favourably with large-scale farmers in the international food trade, their successful transition into market-oriented agriculture will require the adoption of appropriate innovations in food quality and safety management, including traceability. Where opportunities exist for export or regional marketing of products to a major retailer, this will require partnerships to guarantee the quality and traceability of products. With an increasing proportion of the consuming class expected to spend more on high quality products, growers/farmers in these developing regions need access to appropriate quality management tools to assure the quality and traceability of their products.

Food policy implications

It has been reported recently that one of the major factors limiting the marketing potential of fresh commodities produced by small-scale farmers in developing countries is the low confidence that suppliers and importers have in their product quality and safety.[26] Traceability must therefore be incorporated into development projects aimed at improving the quality management systems of such small- and medium-scale rural farmers. As Dolan and Humphrey demonstrated in their study of fresh vegetable exports to the European Union from Africa, excluding small-scale farmers from global food trade can only be reversed if quality, consistency and safety standards are met.

In order to include traceability as a quality index and criterion for food trade, constraints related to increased investment in information technology, data acquisition, storage and retrieval would need to be overcome. Both the exporting and importing countries would also have to implement appropriate trade and economic policies to facilitate trade under such increasingly stringent conditions. Other policy and institutional concerns related to traceability in food trade include providing technical support for training and capacity building, the harmonization of traceability standards with existing food quality standards, and codes of conduct addressing the legal aspects of a breach. Such a legal framework must ensure that food traceability is not used improperly as a barrier to fair trade and for victimization.

One of the consequences of the diversity of international food supply chain practices is that it is critical to adopt a holistic approach to food traceability. This is only achievable by developing and deploying international standards on traceability. There is also a need to harmonize the regulatory framework that will facilitate the identification of fresh and processed food from field to plate.

Conclusions

The globalization of the world economy and freer trade in agricultural products have increased competition, as well as enhancing the flow of food products across national and regional borders. The emergence of multi- and transnational food corporations has resulted in increased outsourcing of products from diverse geographical origins and growing conditions. The eating and shopping attitudes of consumers are also changing with convenience having a major impact on food choices. In addition to product price, consumers are much more attentive and interested in several other factors such as food growing practices, the source of inputs such as seeds, safety and health benefits, and protection of the environment.

Interest in traceability has risen steadily in response to declining public confidence in the ability of current food supply systems to guarantee food safety and other ethical considerations. Recent food scares and the incidence of illnesses related to food poisoning have exacerbated the problem. The high publicity accorded to outbreaks of food-borne diseases has brought food quality and safety under considerable public scrutiny. Use of GM materials, hormones and other growth regulators, animal welfare issues, sociocultural food preferences, production methods, sustainability of agricultural practices and environmental protection are some of the factors influencing consumer perceptions of food quality and safety.

'Traceability' is the term that describes the ability to identify the location and history of a product using recorded data. If properly applied, traceability is an essential part of a good quality assurance system. It facilitates reassurance and enhances customer confidence through effective determination of the origin of food quality and safety problems. It also provides an essential communication linkage between the various players in the food chain, and thereby represents a precautionary approach to food quality management. Traceability offers a win–win opportunity for everyone through collaborative and cooperative responsibility for ensuring food safety and wholesomeness.

Successful implementation of traceability as part of the overall quality and safety management strategy requires training and investments in both human resources and the relevant information and communication technology. This presents a major challenge for small-scale subsistence farmers who dominate agriculture in most developing economies, where basic infrastructure is highly underdeveloped. Resource-poor farmers will need considerable assistance to access and apply these enabling technologies to make it possible for them successfully to transfer to market-oriented agriculture. A new policy initiative is warranted to incorporate traceability specifically as part of the overall quality assurance system in food trade.

Development of international food traceability standards and harmonization of the regulatory framework are critical in view of the diversity of international food supply chain practices.

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